Appendix

Kyrgyz Republic: Debt Sustainability (DSA)—Update

In view of the Kyrgyz Republic's request for access to Fund resources under the Rapid Credit Facility, Fund staff has updated Kyrgyz Republic's debt sustainability assessment.

1. **The previous DSA presented to the Executive Board found that the Kyrgyz Republic faced a moderate risk of debt distress.** The current assessment updates that joint Fund/World Bank analysis. It incorporates the planned large fiscal expansion as part of the authorities' crisis response in the aftermath of the April and June 2010 events.[1] Additionally, it includes the projected use of Fund resources (about US$34 million or SDR 22.2 million) as well as the planned large scale investments in the energy sector that are to be financed primarily through external borrowing on concessional terms (about US$550 million over the next three years).[2] Debt stocks for 2009 have been updated. Finally, the macroeconomic assumptions underlying the DSA have also been revised, mainly to reflect the higher imports (linked to investments) over the medium term, and the long-term impact of the global crisis on remittances levels.

2. **Staff is of the view that the Kyrgyz Republic continues to face a moderate risk of debt distress.** While debt sustainability ratios will deteriorate in the near term due to the expected significant increase in borrowing (Figure 1), the debt outlook is not expected to deteriorate significantly in the longer term given the preexisting borrowing space. While one of the key debt ratios (PV of external debt to GDP) is breached under the baseline scenario starting in 2010, the breach is temporary (as the ratio falls below the 40 percent threshold after 2017) and is not large (it peaks at 48 percent of GDP). Furthermore, when workers' remittances are included in the determination of indicative debt thresholds, all debt ratios are below the modified thresholds (Figure 2).[3] The use of Fund resources under the RCF will have a negligible impact on the debt ratios. The results of the stress tests continue to show a vulnerability to a combination of exogenous shocks. This warrants a prudent approach to the contracting of new external debt, focusing on concessional financing to support the country's large development needs.

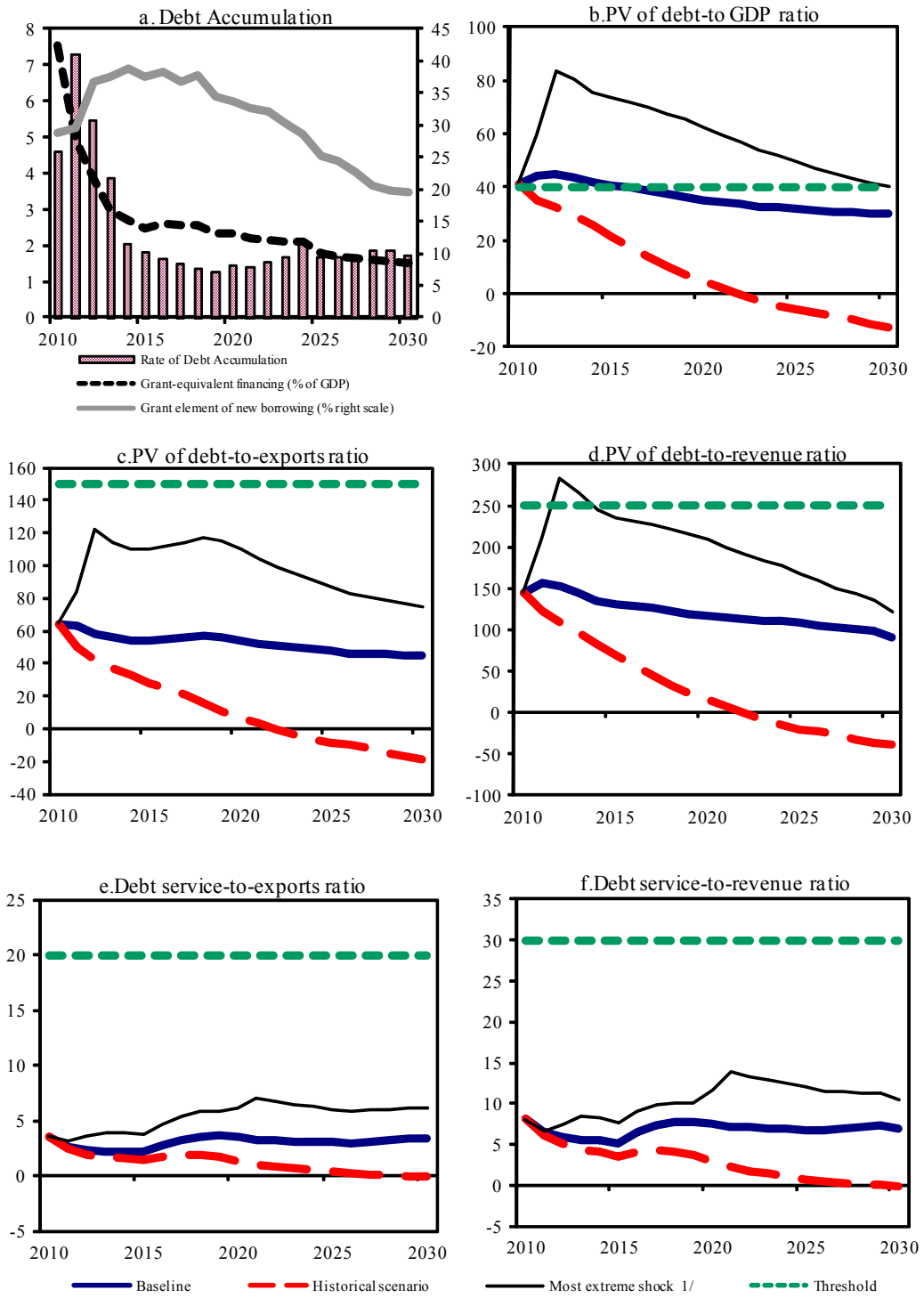
---

[1] Of the US$253 million external financing of the budget deficit in 2010, about US$180 million is expected in concessional loans. The deficit includes the cost of an expected one-time capitalization of AUB (about 0.8 percent of GDP).

[2] The planned investments in the energy sector have been scaled down from US$850 million to US$550 million out of debt sustainability concerns.

[3] As per the *Staff Guidance Note on the Application of the Joint Bank-Fund Debt Sustainability Framework for Low-Income Countries of January 2010*, remittances are included in the denominator of the modified debt burden ratios because they are an important and reliable source of foreign exchange.

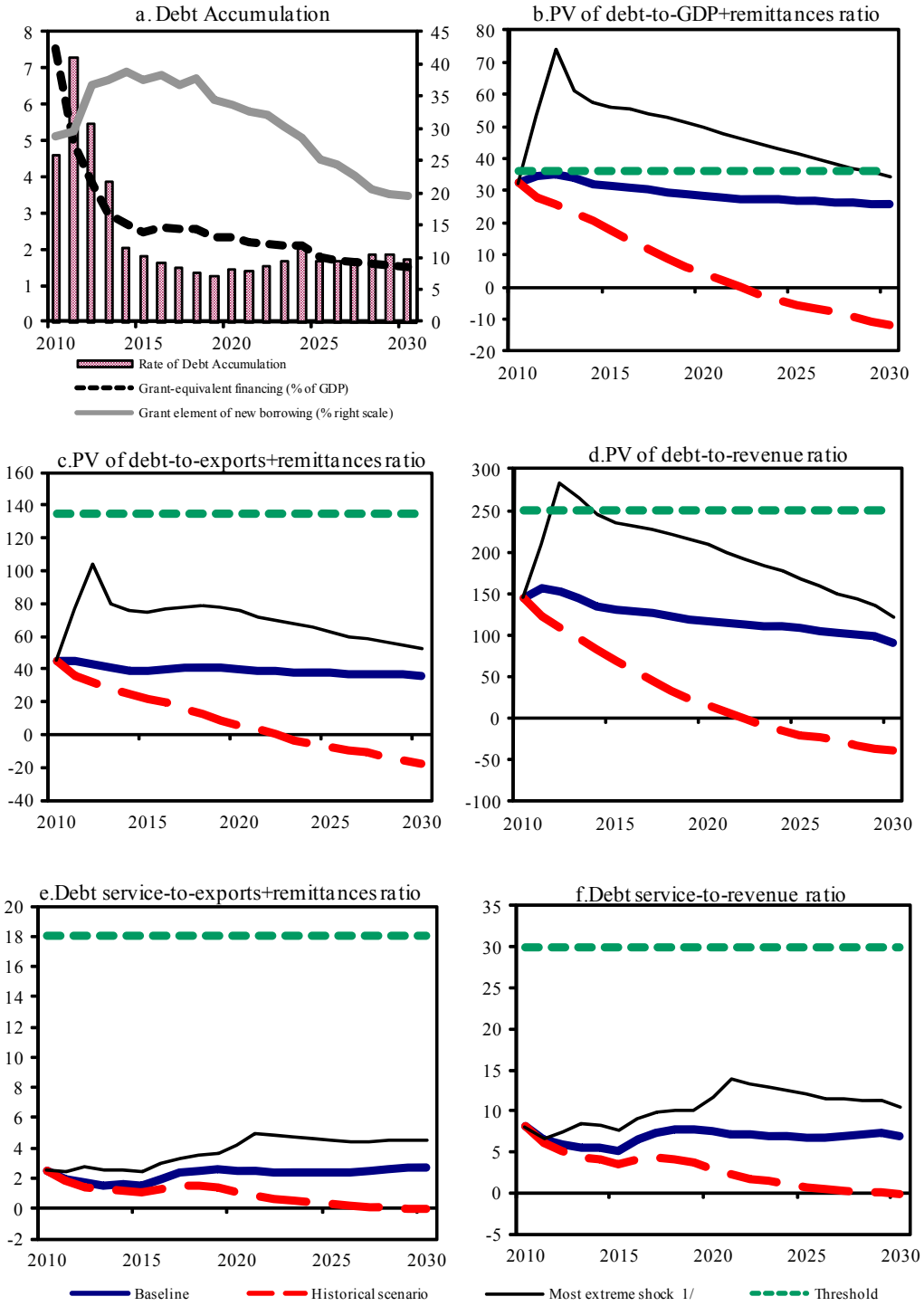
Figure 1. Kyrgyz Republic: Indicators of Public and Publicly Guaranteed External Debt under Alternatives Scenarios, 2010-2030 1/



Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in 2020. In figure b. it corresponds to a Combination shock; in c. to a Exports shock; in d. to a Combination shock; in e. to a Exports shock and in figure f. to a Combination shock

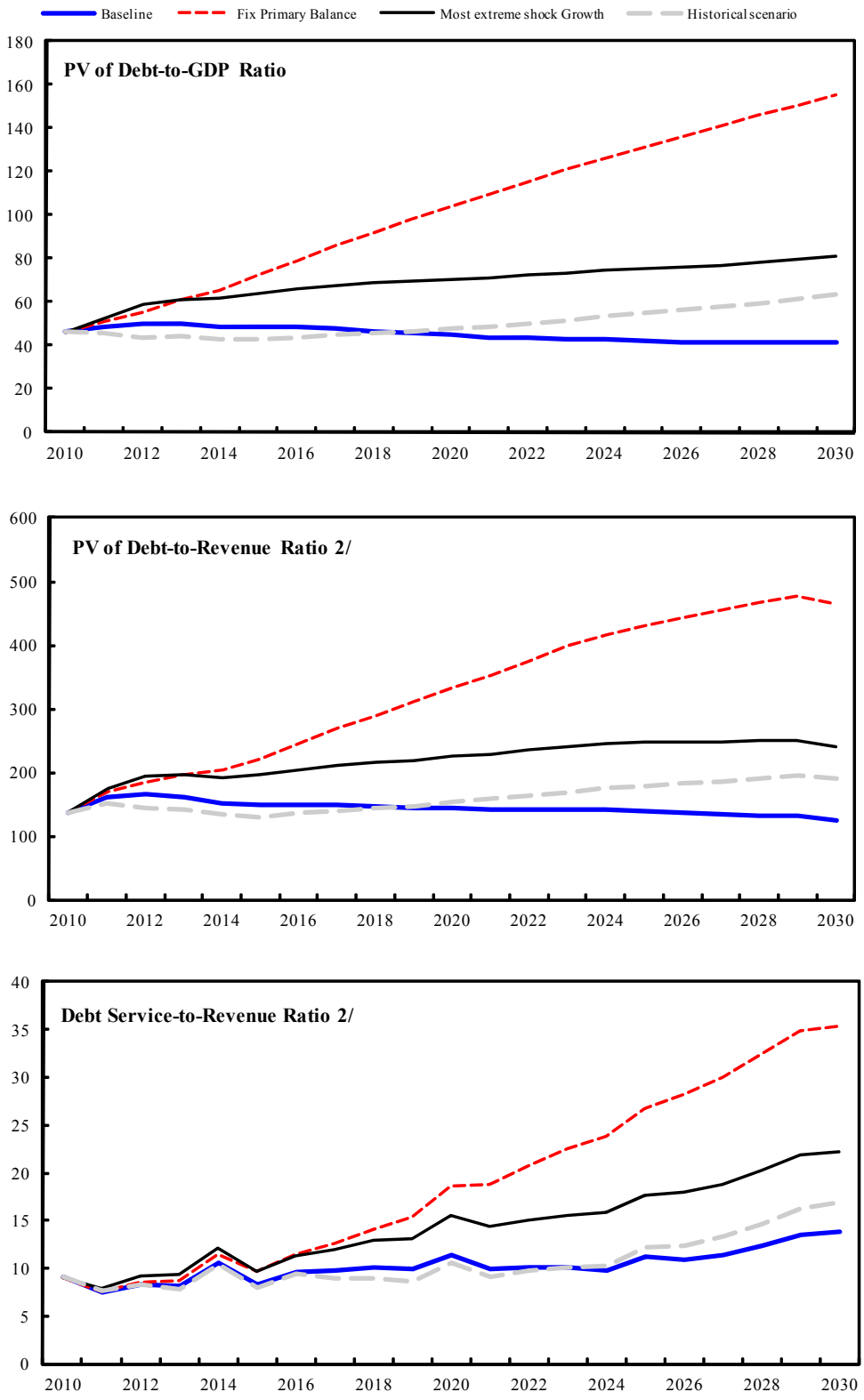
Figure 2. Kyrgyz Republic: Indicators of Public and Publicly Guaranteed External Debt under Alternatives Scenarios Including Remittances, 2010-2030



Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in 2020. In figure b. it corresponds to a Combination shock; in c. to a Combination shock; in d. to a Combination shock; in e. to a Combination shock and in figure f. to a Combination shock

Figure 3. Kyrgyz Republic: Indicators of Public Debt Under Alternative Scenarios, 2010-2030 1/



Sources: Country authorities; and staff estimates and projections.
1/ The most extreme stress test is the test that yields the highest ratio in 2020.
2/ Revenues are defined inclusive of grants.

Table 1a.Kyrgyz Republic: Public Sector Debt Sustainability Framework, Baseline Scenario, 2007-2030
(In percent of GDP, unless otherwise indicated)

	Actual					Estimate	Projections								
	2007	2008	2009	Average	Standard Deviation	2010	2011	2012	2013	2014	2015	2010-15 Average	2020	2030	2016-30 Average
Public sector debt 1/	56.8	48.5	59.4			70.0	68.2	69.2	70.0	67.7	67.5		55.9	40.9	
o/w foreign-currency denominated	52.9	45.6	56.5			66.2	64.4	64.9	63.5	59.6	57.9		49.3	38.9	
Change in public sector debt	-15.7	-8.3	10.9			10.7	-1.8	1.0	0.8	-2.3	-0.1		-2.5	-0.5	
Identified debt-creating flows	-17.9	-9.5	6.9			14.0	1.2	0.4	-1.4	-2.2	0.0		-4.2	-4.2	
Primary deficit	0.0	-1.4	3.0	2.5	2.4	10.9	7.6	6.6	4.3	3.6	3.1	6.0	-1.0	-2.1	-1.1
Revenue and grants	30.3	29.9	33.0			33.9	30.3	30.2	30.9	31.8	32.2		32.7	34.8	
of which: grants	2.3	1.9	5.3			5.3	1.9	0.8	0.8	1.5	1.4		1.2	0.8	
Primary (noninterest) expenditure	30.3	28.5	36.0			44.8	37.9	36.7	35.2	35.5	35.3		31.6	32.7	
Automatic debt dynamics	-17.6	-8.2	3.9			4.2	-6.4	-6.1	-5.7	-5.8	-3.0		-3.1	-2.1	
Contribution from interest rate/growth differential	-7.1	-5.3	-0.6			2.7	-4.4	-4.1	-4.2	-4.1	-2.4		-3.1	-2.1	
of which: contribution from average real interest rate	-1.4	-0.9	0.4			0.6	0.2	-0.2	-0.4	-0.4	-0.4		-0.3	0.0	
of which: contribution from real GDP growth	-5.7	-4.4	-1.1			2.1	-4.6	-3.9	-3.8	-3.7	-2.0		-2.8	-2.0	
Contribution from real exchange rate depreciation	-10.5	-3.0	4.5			1.5	-2.0	-2.0	-1.4	-1.7	-0.7		...	...	
Other identified debt-creating flows	-0.2	0.1	0.0			-1.1	-0.1	-0.1	-0.1	-0.1	-0.1		0.0	0.0	
Privatization receipts (negative)	-0.1	0.1	0.0			-1.1	-0.1	-0.1	-0.1	-0.1	-0.1		0.0	0.0	
Recognition of implicit or contingent liabilities	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Debt relief (HIPC and other)	-0.1	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Other (specify, e.g. bank recapitalization)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Residual, including asset changes	2.2	1.1	4.0			-3.4	-3.0	0.6	2.2	-0.1	-0.1		1.7	3.7	
Other Sustainability Indicators															
PV of public sector debt	5.1	4.0	38.4			46.3	48.8	49.7	51.0	50.2	50.9		42.1	32.0	
o/w foreign-currency denominated	1.2	1.0	35.6			42.5	45.0	45.4	44.6	42.1	41.3		35.5	30.0	
o/w external	...	...	34.5			41.5	44.2	44.7	44.0	41.5	40.8		35.2	29.8	
PV of contingent liabilities (not included in public sector debt)	...	...	...			...	...	...	...	...	...		...	...	
Gross financing need 2/	3.2	1.4	6.8			15.4	11.7	10.7	9.4	10.9	9.8		4.9	1.0	
PV of public sector debt-to-revenue and grants ratio (in percent)	16.9	13.2	116.3			136.6	161.2	164.6	165.1	157.7	158.0		128.9	91.9	
PV of public sector debt-to-revenue ratio (in percent)	18.3	14.1	138.6			162.0	172.1	169.4	169.5	165.3	165.2		133.7	94.1	
o/w external 3/	…	…	124.6			145.1	155.8	152.4	146.1	136.8	132.4		111.8	87.8	
Debt service-to-revenue and grants ratio (in percent) 4/	6.4	5.5	7.3			9.3	7.6	8.1	9.9	12.7	10.1		8.4	8.2	
Debt service-to-revenue ratio (in percent) 4/	7.0	5.9	8.7			11.0	8.1	8.3	10.2	13.3	10.5		8.7	8.4	
Primary deficit that stabilizes the debt-to-GDP ratio	15.7	7.0	-7.9			0.2	9.5	5.6	3.5	5.9	3.2		1.5	-1.6	
Key macroeconomic and fiscal assumptions															
Real GDP growth (in percent)	8.5	8.4	2.3	4.8	3.4	-3.5	7.1	6.1	5.9	5.6	3.0	4.0	5.1	5.1	5.0
Average nominal interest rate on forex debt (in percent)	0.9	0.9	1.2	1.5	0.5	1.3	1.0	1.1	1.2	1.2	1.2	1.2	1.3	1.8	1.4
Average real interest rate on domestic debt (in percent)	-7.3	-7.0	12.9	-2.1	8.2	4.0	1.6	1.6	1.2	0.7	0.4	1.6	0.2	0.3	0.3
Real exchange rate depreciation (in percent, + indicates depreciation)	-17.2	-6.2	10.1	-7.1	7.8	2.6	...	...	...	...	...	...	...	...	...
Inflation rate (GDP deflator, in percent)	14.9	22.2	2.1	10.4	9.2	9.3	7.1	8.1	7.5	8.1	6.2	7.7	4.0	4.0	4.1
Growth of real primary spending (deflated by GDP deflator, in percent)	0.2	0.0	0.3	0.1	0.1	0.2	-0.1	0.0	0.0	0.1	0.0	0.0	0.0	0.1	0.0
Grant element of new external borrowing (in percent)	...	...	...	…	…	28.7	29.6	36.9	37.6	38.8	37.6	34.9	33.6	19.5	...

Sources: Country authorities; and staff estimates and projections.
1/ General government gross debt.
2/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.
3/ Revenues excluding grants.
4/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.
5/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

Table 2a.Kyrgyz Republic: Sensitivity Analysis for Key Indicators of Public Debt 2010-2030

	Projections							
	2010	2011	2012	2013	2014	2015	2020	2030
PV of Debt-to-GDP Ratio								
Baseline	46	49	50	50	48	48	45	42
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	46	46	44	44	43	43	48	64
A2. Primary balance is unchanged from 2010	46	51	56	61	66	72	104	155
A3. Permanently lower GDP growth 1/	46	49	51	52	51	53	57	80
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2011-2012	46	53	59	61	62	64	70	81
B2. Primary balance is at historical average minus one standard deviations in 2011-2012	46	47	46	47	45	45	42	40
B3. Combination of B1-B2 using one half standard deviation shocks	46	47	47	49	50	51	55	64
B4. One-time 30 percent real depreciation in 2011	46	67	66	64	61	61	55	50
B5. 10 percent of GDP increase in other debt-creating flows in 2011	46	57	57	57	55	55	50	45
PV of Debt-to-Revenue Ratio 2/								
Baseline	137	161	165	161	151	148	144	124
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	137	150	145	141	133	130	153	190
A2. Primary balance is unchanged from 2010	137	170	184	197	204	221	334	464
A3. Permanently lower GDP growth 1/	137	163	169	167	160	161	183	237
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2011-2012	137	174	194	197	191	195	225	241
B2. Primary balance is at historical average minus one standard deviations in 2011-2012	137	154	153	150	141	139	136	120
B3. Combination of B1-B2 using one half standard deviation shocks	137	156	155	158	154	157	178	190
B4. One-time 30 percent real depreciation in 2011	137	221	217	206	191	186	176	149
B5. 10 percent of GDP increase in other debt-creating flows in 2011	137	188	190	184	171	168	161	134
Debt Service-to-Revenue Ratio 2/								
Baseline	9	8	8	8	11	8	11	14
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	9	8	8	8	10	8	11	17
A2. Primary balance is unchanged from 2010	9	8	9	9	11	10	19	35
A3. Permanently lower GDP growth 1/	9	8	8	8	11	9	13	21
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2011-2012	9	8	9	9	12	10	16	22
B2. Primary balance is at historical average minus one standard deviations in 2011-2012	9	8	8	8	10	8	11	13
B3. Combination of B1-B2 using one half standard deviation shocks	9	8	9	8	11	9	12	18
B4. One-time 30 percent real depreciation in 2011	9	9	11	11	13	11	15	19
B5. 10 percent of GDP increase in other debt-creating flows in 2011	9	8	9	9	11	9	13	15

Sources: Country authorities; and staff estimates and projections.
1/ Assumes that real GDP growth is at baseline minus one standard deviation divided by the square root of the length of the projection period.
2/ Revenues are defined inclusive of grants.

Table 3a.: External Debt Sustainability Framework, Baseline Scenario, 2007-2030 1/
(In percent of GDP, unless otherwise indicated)

	Actual			Historical 0 Average	Standard 0 Deviation	Projections									
	2007	2008	2009			2010	2011	2012	2013	2014	2015	2010-2015 Average	2020	2030	2016-2030 Average
External debt (nominal) 1/	**57.2**	**48.4**	**61.0**			**70.9**	**67.2**	**66.3**	**64.5**	**60.5**	**58.9**		**50.0**	**39.3**	
o/w public and publicly guaranteed (PPG)	51.7	44.5	55.5			65.2	63.6	64.2	62.9	59.0	57.4		49.0	38.7	
Change in external debt	-16.9	-8.9	12.6			9.9	-3.7	-0.9	-1.8	-3.9	-1.6		-1.7	-0.9	
Identified net debt-creating flows	-24.2	-11.9	-0.2			4.3	0.8	-2.2	-3.1	-3.9	-1.4		-2.1	-0.2	
Non-interest current account deficit	**-0.5**	**7.5**	**-3.1**	**-0.6**	**4.0**	**3.8**	**8.1**	**4.2**	**3.3**	**2.5**	**2.4**		**2.6**	**3.2**	3.0
Deficit in balance of goods and services	25.6	33.3	18.8			28.2	34.4	26.7	26.2	24.3	26.3		25.8	18.9	
Exports	59.0	59.2	61.7			64.7	70.3	76.6	79.3	77.5	76.1		65.2	66.8	
Imports	84.6	92.5	80.5			92.9	104.7	103.3	105.5	101.8	102.4		90.9	85.7	
Net current transfers (negative = inflow)	-26.8	-28.8	-26.7	-17.0	9.7	-28.8	-28.6	-28.2	-28.8	-29.1	-29.1		-23.9	-16.2	-21.0
o/w official	-0.8	-0.9	-4.5			-1.4	-0.6	-0.2	-0.2	-0.2	-0.2		-0.1	-0.1	
Other current account flows (negative = net inflow)	0.7	2.9	4.8			4.5	2.2	5.6	5.9	7.2	5.3		0.7	0.5	
Net FDI (negative = inflow)	**-5.5**	**-5.2**	**-4.2**	**-3.1**	**2.7**	**-3.2**	**-3.9**	**-4.0**	**-4.0**	**-4.3**	**-3.3**		**-3.2**	**-2.6**	-3.0
Endogenous debt dynamics 2/	**-18.2**	**-14.3**	**7.0**			**3.7**	**-3.4**	**-2.4**	**-2.4**	**-2.1**	**-0.6**		**-1.4**	**-0.8**	
Contribution from nominal interest rate	0.6	0.5	1.0			1.6	1.3	1.3	1.2	1.1	1.1		1.0	1.1	
Contribution from real GDP growth	-4.7	-3.6	-1.2			2.2	-4.7	-3.7	-3.5	-3.3	-1.7		-2.4	-1.9	
Contribution from price and exchange rate changes	-14.2	-11.2	7.2			…	…	…	…	…	…		…	…	
Residual (3-4) 3/	**7.3**	**3.1**	**12.8**			**5.6**	**-4.6**	**1.4**	**1.2**	**0.0**	**-0.2**		**0.4**	**-0.7**	
o/w exceptional financing	-0.1	-0.1	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
PV of external debt 4/	...	...	40.0			47.2	47.8	46.8	45.5	43.1	42.3		36.3	30.4	
In percent of exports	...	...	64.9			72.9	67.9	61.1	57.5	55.6	55.5		55.6	45.4	
PV of PPG external debt	**...**	**...**	**34.5**			**41.5**	**44.2**	**44.7**	**44.0**	**41.5**	**40.8**		**35.2**	**29.8**	
In percent of exports	**...**	**...**	**56.0**			**64.2**	**62.8**	**58.3**	**55.5**	**53.6**	**53.6**		**54.0**	**44.6**	
In percent of government revenues	**...**	**...**	**124.6**			**145.1**	**155.8**	**151.9**	**145.4**	**135.8**	**130.6**		**117.8**	**91.3**	
Debt service-to-exports ratio (in percent)	**4.5**	**3.6**	**6.9**			**6.9**	**6.0**	**5.2**	**3.8**	**3.2**	**3.2**		**4.6**	**4.3**	
PPG debt service-to-exports ratio (in percent)	**3.1**	**2.5**	**3.2**			**3.6**	**2.7**	**2.3**	**2.1**	**2.2**	**2.1**		**3.4**	**3.4**	
PPG debt service-to-revenue ratio (in percent)	**6.5**	**5.2**	**7.1**			**8.0**	**6.6**	**6.0**	**5.5**	**5.5**	**5.2**		**7.5**	**7.0**	
Total gross financing need (Billions of U.S. dollars)	-0.1	0.2	-0.1			0.2	0.4	0.2	0.1	0.0	0.1		0.2	0.7	
Non-interest current account deficit that stabilizes debt ratio	16.4	16.3	-15.6			-6.1	11.8	5.0	5.1	6.4	4.0		4.3	4.1	
Key macroeconomic assumptions															
Real GDP growth (in percent)	8.5	8.4	2.3	4.8	3.4	-3.5	7.1	6.1	5.9	5.6	3.0	4.0	5.1	5.1	5.0
GDP deflator in US dollar terms (change in percent)	23.6	24.4	-12.9	9.4	10.5	0.7	0.1	4.9	4.4	4.9	3.2	3.0	2.0	2.0	1.9
Effective interest rate (percent) 5/	1.2	1.3	1.9	1.7	0.6	2.5	2.0	2.1	1.9	2.0	2.0	2.1	2.1	2.9	2.3
Growth of exports of G&S (US dollar terms, in percent)	51.1	35.4	-7.1	19.7	19.5	1.9	16.5	21.3	14.4	8.3	4.5	11.1	7.9	7.0	6.1
Growth of imports of G&S (US dollar terms, in percent)	42.8	47.5	-22.5	20.9	27.0	12.2	20.8	9.8	12.8	7.0	6.9	11.6	7.3	6.8	5.7
Grant element of new public sector borrowing (in percent)	...	...	...	...	...	28.7	29.6	36.9	37.6	38.8	37.6	34.9	33.6	19.5	29.1
Government revenues (excluding grants, in percent of GDP)	28.1	28.0	27.7			28.6	28.4	29.4	30.3	30.6	31.2		29.9	32.7	30.2
Aid flows (in Billions of US dollars) 7/	0.2	0.1	0.6			0.3	0.2	0.3	0.2	0.2	0.2		0.3	0.3	
o/w Grants	0.1	0.1	0.2			0.2	0.1	0.0	0.0	0.1	0.1		0.1	0.2	
o/w Concessional loans	0.1	0.0	0.4			0.1	0.1	0.3	0.2	0.1	0.1		0.1	0.1	
Grant-equivalent financing (in percent of GDP) 8/	...	...	...			7.6	5.0	3.8	3.0	2.7	2.5		2.3	1.5	2.0
Grant-equivalent financing (in percent of external financing) 8/	...	...	...			57.7	40.7	42.9	45.2	58.0	58.5		50.8	35.2	46.1
Memorandum items:															
Nominal GDP (Billions of US dollars)	3.8	5.1	4.6			4.4	4.8	5.3	5.9	6.5	6.9		9.6	19.0	
Nominal dollar GDP growth	34.1	34.8	-10.9			-2.8	7.2	11.3	10.5	10.8	6.3	7.2	7.1	7.2	7.0
PV of PPG external debt (in Billions of US dollars)			1.5			1.7	2.1	2.3	2.5	2.7	2.8		3.3	5.6	
(PVt-PVt-1)/GDPt-1 (in percent)						4.6	7.3	5.5	3.9	2.0	1.8	4.2	1.4	1.7	1.6
Gross remittances (Billions of US dollars)	1.0	1.4	1.0			1.2	1.3	1.5	1.7	1.9	2.0		2.3	3.1	
PV of PPG external debt (in percent of GDP + remittances)	...	...	28.3			32.6	34.5	34.9	34.2	32.2	31.6		28.4	25.7	
PV of PPG external debt (in percent of exports + remittances)	...	...	41.2			45.0	44.9	42.7	40.8	39.1	38.8		39.6	35.9	
Debt service of PPG external debt (in percent of exports + remittances)	...	...	2.3			2.5	1.9	1.7	1.6	1.6	1.5		2.5	2.8	

Sources: Country authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt.
2/ Derived as [r - g - ρ(1+g)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.
3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.
4/ Assumes that PV of private sector debt is equivalent to its face value.
5/ Current-year interest payments divided by previous period debt stock.
6/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.
7/ Defined as grants, concessional loans, and debt relief.
8/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

Table 3b.Kyrgyz Republic: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2010-2030
(In percent)

	Projections							
	2010	2011	2012	2013	2014	2015	**2020**	2030
PV of debt-to GDP ratio								
Baseline	42	44	45	44	42	41	**35**	30
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2010-2030 1/	42	35	32	29	26	21	**5**	-12
A2. New public sector loans on less favorable terms in 2010-2030 2	42	46	49	49	47	47	**45**	46
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2011-2012	42	46	49	48	45	44	**38**	32
B2. Export value growth at historical average minus one standard deviation in 2011-2012 3/	42	51	67	64	61	59	**51**	35
B3. US dollar GDP deflator at historical average minus one standard deviation in 2011-2012	42	44	47	47	44	43	**37**	32
B4. Net non-debt creating flows at historical average minus one standard deviation in 2011-2012 4/	42	61	78	75	71	69	**59**	38
B5. Combination of B1-B4 using one-half standard deviation shocks	42	60	83	80	75	74	**63**	40
B6. One-time 30 percent nominal depreciation relative to the baseline in 2011 5/	42	62	63	62	58	57	**50**	42
PV of debt-to-exports ratio								
Baseline	64	63	58	55	54	54	**54**	45
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2010-2030 1/	64	50	42	37	33	28	**7**	-19
A2. New public sector loans on less favorable terms in 2010-2030 2	64	65	64	62	61	62	**70**	69
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2011-2012	64	62	57	55	53	53	**53**	44
B2. Export value growth at historical average minus one standard deviation in 2011-2012 3/	64	84	122	114	110	110	**110**	74
B3. US dollar GDP deflator at historical average minus one standard deviation in 2011-2012	64	62	57	55	53	53	**53**	44
B4. Net non-debt creating flows at historical average minus one standard deviation in 2011-2012 4/	64	87	102	95	91	91	**90**	57
B5. Combination of B1-B4 using one-half standard deviation shocks	64	90	123	114	110	109	**109**	68
B6. One-time 30 percent nominal depreciation relative to the baseline in 2011 5/	64	62	57	55	53	53	**53**	44
PV of debt-to-revenue ratio								
Baseline	145	156	152	145	136	131	**118**	91
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2010-2030 1/	145	123	110	96	83	68	**15**	-38
A2. New public sector loans on less favorable terms in 2010-2030 2	145	162	165	163	155	151	**152**	140
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2011-2012	145	162	165	158	148	142	**128**	99
B2. Export value growth at historical average minus one standard deviation in 2011-2012 3/	145	178	226	213	198	190	**170**	108
B3. US dollar GDP deflator at historical average minus one standard deviation in 2011-2012	145	155	161	154	144	138	**125**	97
B4. Net non-debt creating flows at historical average minus one standard deviation in 2011-2012 4/	145	216	266	249	231	221	**197**	116
B5. Combination of B1-B4 using one-half standard deviation shocks	145	210	284	265	246	236	**210**	123
B6. One-time 30 percent nominal depreciation relative to the baseline in 2011 5/	145	219	213	204	191	183	**166**	129

Table 3b.Kyrgyz Republic: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2010-2030 (continued)
(In percent)

Debt service-to-exports ratio

Baseline	4	3	2	2	2	2	**3**	3
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2010-2030 1/	4	2	2	2	2	1	**1**	0
A2. New public sector loans on less favorable terms in 2010-2030 2	4	3	2	2	3	3	**4**	4
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2011-2012	4	3	2	2	2	2	**3**	3
B2. Export value growth at historical average minus one standard deviation in 2011-2012 3/	4	3	4	4	4	4	**6**	6
B3. US dollar GDP deflator at historical average minus one standard deviation in 2011-2012	4	3	2	2	2	2	**3**	3
B4. Net non-debt creating flows at historical average minus one standard deviation in 2011-2012 4/	4	3	3	3	3	3	**5**	5
B5. Combination of B1-B4 using one-half standard deviation shocks	4	3	3	4	4	4	**6**	6
B6. One-time 30 percent nominal depreciation relative to the baseline in 2011 5/	4	3	2	2	2	2	**3**	3

Debt service-to-revenue ratio

Baseline	8	7	6	6	6	5	**8**	7
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2010-2030 1/	8	6	5	4	4	3	**3**	0
A2. New public sector loans on less favorable terms in 2010-2030 2	8	7	6	6	7	6	**9**	9
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2011-2012	8	7	7	6	6	6	**8**	8
B2. Export value growth at historical average minus one standard deviation in 2011-2012 3/	8	7	7	7	7	7	**9**	9
B3. US dollar GDP deflator at historical average minus one standard deviation in 2011-2012	8	7	6	6	6	6	**8**	8
B4. Net non-debt creating flows at historical average minus one standard deviation in 2011-2012 4/	8	7	7	8	8	7	**11**	10
B5. Combination of B1-B4 using one-half standard deviation shocks	8	7	7	9	8	8	**12**	10
B6. One-time 30 percent nominal depreciation relative to the baseline in 2011 5/	8	9	9	8	8	7	**11**	10
Memorandum item:								
Grant element assumed on residual financing (i.e., financing required above baseline) 6/	28	28	28	28	28	28	**28**	28

Sources: Country authorities; and staff estimates and projections.

1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
2/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline., while grace and maturity periods are the same as in the baseline.
3/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly assuming an offsetting adjustment in import levels).
4/ Includes official and private transfers and FDI.
5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.
6/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.